

Mid-term examination on Algebraic Specification

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Question 1. We want an algebraic specification of sets made of elements of the same type `element`. Let us call it `SET(element)`. Here is the signature:

- **Defined types**

- The type of the sets is noted `t`.
- The type of the set elements is noted `element`.

- **Constructors**

- `EMPTY : t`
The term `EMPTY` represents the set (noted $\emptyset$ in mathematics) which contains no element.
- `ADD : element × t → t`
The term `ADD(e, s)` denotes the set `s` augmented by adding element `e`. If `e` was already in `s`, then the result is `s`. The order in which elements are added is not meaningful, so all function on sets should not depend on any element order.

- **Functions**

- `UNION : t × t → t`
The term `UNION(s1, s2)` denotes the set made of the elements of `s1` and `s2` (noted $s_1 \cup s_2$ in mathematics). It is commutative and associative.
- `INTER : t × t → t`
The term `INTER(s1, s2)` denotes the set made of elements of `s1` which also belong to `s2` (noted $s_1 \cap s_2$ in mathematics). The function name stands for “intersection”. It is commutative and associative. It is distributive over the set union.
- `MEM : element × t → boolean`
The term `MEM(e, s)` is **true** if and only if `e` belongs to `s`, otherwise it is **false**. The function name is a shorthand for “membership”.

Complete this signature with equations defining the constructors and functions.

Question 2. Consider the $\text{STACK}(\text{item})$ signature as follows:

- **Defined types**

- The type of the stacks is called $\mathbf{t}$.
- The type of the items is called $\mathbf{item}$.

- **Constructors**

- $\text{EMPTY} : \mathbf{t}$
Term EMPTY denotes the empty stack.
- $\text{PUSH} : \mathbf{item} \times \mathbf{t} \rightarrow \mathbf{t}$
Term $\text{PUSH}(e, s)$ denotes stack s with item e on the top.

- **Functions**

- $\text{POP} : \mathbf{t} \rightarrow \mathbf{item} \times \mathbf{t}$
This is the projection of PUSH .
- $\text{APPEND} : \mathbf{t} \times \mathbf{t} \rightarrow \mathbf{t}$
Term $\text{APPEND}(s_1, s_2)$ denotes the stack made of s_2 on the bottom and s_1 on the top.
- $\text{NTH} : \mathbf{t} \times \mathbf{int} \rightarrow \mathbf{item}$
Term $\text{NTH}(s, n)$ denotes the n^{th} item in stack s . The item must exist.
- $\text{FLATTEN} : \text{STACK}(\text{STACK}(\mathbf{item}).\mathbf{t}).\mathbf{t} \rightarrow \text{STACK}(\mathbf{item})$
Term $\text{FLATTEN}(s)$, where s denotes a stack of stack of items, represents the stack made by appending all the stacks in s , in the same order.
- $\text{EXISTS} : (\mathbf{item} \rightarrow \mathbf{boolean}) \times \text{STACK}(\mathbf{item}).\mathbf{t} \rightarrow \mathbf{boolean}$
Term $\text{EXISTS}(f, s)$ is **true** if and only if there exists an item e in s such that $f(e)$ is **true**. Otherwise it is **false**.

Give equations defining NTH , FLATTEN and EXISTS .